

Program 2

Write an 8051 program to transfer a string of data from code space starting at address 200H to RAM locations starting at 40H. The data is as shown below:

0200H:DB "VIT UNIVERSITY"

Using the simulator, single-step through the program and examine the data transfer and registers.

```
ORG 0000H

MOV A,#00H

MOV DPTR,#0200H

MOV R1,#0EH

MOV R0,#40H

LOOP: CLR A

      MOVC A,@A+DPTR

      MOV @R0,A

      INC DPTR

      INC R0

      DJNZ R1,LOOP

      HERE: SJMP HERE


ORG 0200H

DB "VIT UNIVERSITY"

END
```